

An Analysis of the Factors Which Determine Signal/Noise Discrimination in Pulsed-Carrier Systems*

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Summary—The smallest signal discernible through background noise is formulated in terms of the pulse energy, its repetition rate, the receiver design, and the choice of integrating and indicating means.

The smallest signal visible on a Type-A scan can be improved upon by the use of electromechanical integrators. Integration before detection is, in theory, ultimately the most effective, but runs into serious practical difficulties. These are avoided when the integration is performed after detection. Optimum predetector selectivity is formulated. The optimum detector (square law) is found, and other detectors compared. A transmitter criterion provides a basis for comparing the effectiveness of transmitters, and shows how, at the expense of range resolution, longer pulses can increase visibility.

I. ELEMENTS OF THE PROBLEM

A THIN "TARGET" lying within a segment of range Δr reflects a pulsed carrier, producing at the terminals of a receiving antenna an open-circuit voltage,

$$e(t) = F(t) \cos(\omega_0 t + \theta). \quad (1)$$

With a few exceptions, the pulsing function will be supposed rectangular, duration τ , and recurrent, period T , so that in a sample period, $-T/2 < t < T/2$,

$$F(t) = \begin{cases} 0, & -T/2 < t < -\tau/2 \\ V, & -\tau/2 < t < \tau/2 \\ 0, & \tau/2 < t < T/2. \end{cases} \quad (2)$$

The total system noise may be described by a statement of the equivalent open-circuit mean-square noise voltage at the same terminals,

$$\overline{e_n^2} = N(\text{operating}) \cdot 4kT_0 R_a \Delta f \quad (3)$$

where

T_0 = room temperature, degrees Kelvin,

k = Boltzmann's constant,

R_a = antenna resistance,

Δf = predetector noise bandwidth,

$N(\text{operating})$ = operating noise factor [1] of receiver.

(Note: This term refers to total system noise including that arriving via the antenna. It is related to the ordinary noise factor N of the receiver via $N(\text{operating}) \equiv N + T_a/T_0 - 1$, where T_a denotes the noise temperature of the antenna. DON, 1963.)

The problem is simply to discover the conditions which, in view of the noise, permit the presence and absence of the target to be determined to within a specific degree of certainty, that is, to formulate the minimum "visible" signal. The study is, therefore, largely an application of the theory of probabilities. It will be conducted in such a way as to permit a comparative examination of the performance of several integrating (memorizing) schemes designed to lower the minimum visible signal. And, because the memory and judgment of the eye-brain network are hardly susceptible to analytical treatment, in essence the study pertains to purely physical mechanisms whose final function is, let us suppose, to fire or refrain from firing a gun. Even so, the significance of the results will be apparent in situations where observer judgment is a vital component of the receiving mechanism, and one section deals with conventional visual output devices, such as Type-A scan.

In most receivers the noise factor is virtually independent of the design parameters which determine Δf ; consequently, for our purposes the quantity

$$a \equiv N(\text{operating}) \cdot 4kT_0 R_a \quad (4)$$

will be considered a system constant. The quantity

$$W \equiv \int_{-T/2}^{T/2} e^2(t) dt \quad (5)$$

is a measure of the available echo energy contained in one pulse. The dimensionless quotient,

$$\lambda_0 \equiv \frac{W}{a}, \quad (6)$$

being the ratio of pulse energy at the antenna terminals to equivalent noise power per unit bandwidth at the same terminals, will be a much used parameter, since a formulation of the minimum distinguishable value of λ_0 will suffice to define the minimum visible signal.

An unobserved target may hardly be presupposed stationary. However, for simplicity, this will be assumed at first in each analysis. In terms of the framework established, the modifications necessary to accommodate moving targets will then be discussed.

Section II sets forth the necessary theorems in probability together with the statistics of noise and of the envelopes of mixtures of noise plus a sine wave.

Section III examines the possibilities of integration

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before detection. Section IV examines integration after detection. Section V estimates the merits of conventional fluorescent screen and photographic methods.

II. STATISTICS OF NOISE

If x be a chance variable and $p(x)dx$ the probability of finding the variable in the range, $x, x+dx$, the function $p(x)$ will be called the density of probability. The expectation of x (average value) is

$$\bar{x} \equiv \int_{-\infty}^{\infty} xp(x)dx.$$

The dispersion of x (mean-square deviation from the mean) is the expectation of $(x-\bar{x})^2$, thus

$$\overline{(x-\bar{x})^2} \equiv \int_{-\infty}^{\infty} (x-\bar{x})^2 p(x)dx.$$

A. Laplace-Liapounoff Theorem [2]

Let there be $x_1, x_2, \dots, x_n$ independent chance variables each of which is associated with a density of probability, not necessarily the same for each. Let B denote the dispersion of

$$y = \sum_1^n x_k.$$

The theorem states that the density of probability of y , considered as a function of n , has an asymptotic expansion the first term of which is

$$\frac{1}{\sqrt{2\pi B}} e^{-(y-\bar{y})^2/2B} \quad (7)$$

a "normal" distribution. (Note: This theorem is today perhaps more readily identified as one of a vast group of propositions wearing collectively the label: The Central Limit Theorem. DON, 1963.) A proper exposition of the theorem includes a description of the conditions which must be satisfied in order that the asymptotic expansion converge at all as $n \rightarrow \infty$; these can be overlooked without hazard in the applications to follow. It also attempts, at least, to formulate the error involved when, with finite n , only the first term (7) of the expansion is employed. This we have not room to explore, but observe that in the applications we will generally use $n \gg 1$, and remember that (7) becomes more nearly exact for *any* n the more nearly each of the individual densities of probability resembles a "normal" distribution. The latter observation will, in some instances, permit us to use the theorem with negligible error for $n=2$ or 3.

B. Application to Noise Statistics

The above very powerful theorem leads immediately to the, by now, well-known [3] normal distribution for the instantaneous value of noise. Without more ado one concludes that if x be the instantaneous value

of fluctuation current or voltage (dc suppressed) at any point in a *linear* network supplied with any motley assortment whatever of uncorrelated stimuli (independent chance variables) such that the mean-square fluctuation (dispersion) is Z , the density of probability is

$$\frac{1}{\sqrt{2\pi Z}} e^{-x^2/2Z} \quad (8)$$

with an error that vanishes as the peak response to any single stimulus vanishes in comparison with the rms response; in other words, the error decreases the greater the overlap in the wave trains [3]. Network linearity is necessary to preserve independence from source to measuring point. Note the absence of any reference to the frequency spectrum. The formula can certainly be expected to hold for the usual tube and circuit fluctuations, some atmospherics, but generally not for ignition noise (n is too small). In this study it will be supposed that (8) characterizes satisfactorily the noise described by (3).

Now the fluctuation x may, without destroying its chance character, be forecast or reconstructed as a function (but not an analytic function) of the time

$$x = A(t) \cos \omega_0 t + B(t) \sin \omega_0 t, \quad (9)$$

where ω_0 may be chosen altogether arbitrarily, and the independent chance variables, A and B , have densities of probability like that for x , namely

$$\frac{1}{\sqrt{2\pi Z}} e^{-A^2/2Z}, \quad \frac{1}{\sqrt{2\pi Z}} e^{-B^2/2Z} \quad (10)$$

C. Statistics of the Envelope of Noise plus Sine Signal

To define the envelope of a modulated carrier is extremely painful. But in instances where the modulation frequencies are small in comparison with the carrier frequency, one radio engineer's determination of the envelope will agree with another's for practical purposes. This development will consequently be confined to discussion of instances in which an envelope may be said to exist in this sense; that is, to situations in which the noise bandwidth is small relative to (say) the mid-band frequency.

Consider the co-existence, at any point of a linear network, of fluctuations characterized by (9) together with a steady sine signal

$$v \cos (\omega_0 t + \theta),$$

where θ is accidental with density of probability, $1/2\pi$. The functions $A(t)$, $B(t)$ are regarded as slowly varying functions of the time. Hence the envelope of the mixture may be written

$$E = \sqrt{(A + v \cos \theta)^2 + (B - v \sin \theta)^2}$$

With the substitutions

$$E \cos \phi = A + v \cos \theta$$

$$E \sin \phi = B - v \sin \theta$$

and some transformations, we find for E the density of probability

$$P(E) = \frac{E}{Z} e^{-(E^2+v^2)/2Z} I_0\left(\frac{vE}{Z}\right), \quad 0 \leq E, \quad (11)$$

where I_0 is the Bessel function of zero order, imaginary argument. Introducing the parameter

$$\lambda \equiv \frac{v^2}{2Z} \quad (12)$$

to measure the rms value of the sine wave on the rms noise scale, and replacing E by the variable

$$x \equiv \frac{E}{\sqrt{Z}}$$

which measures the envelope against the same scale, we have for x the density of probability¹

$$P(x, \lambda) = x e^{-x^2/2} e^{-\lambda} I_0(\sqrt{2\lambda} x) \quad 0 \leq x \quad (13)$$

$$= x e^{-x^2/2} \left[1 - \lambda \left(1 - \frac{x^2}{2} \right) + O(\lambda^2) \right] \quad (13a)$$

$$\sim \frac{1}{\sqrt{2\pi}} e^{-1/2(x-\sqrt{1+2\lambda})^2} \quad (13b)$$

For $\lambda \ll 1$, the envelope of a mixture of sine signal and noise cannot be distinguished in any way from an increase in the noise itself from Z to $Z(1+\lambda)$. It will be appreciated that the problems of reception to be discussed will, for the most part, be concerned with the detection of λ 's of this magnitude.

In Fig. 1 plots of $P(x)$ for several values of λ are shown. The locus of the peaks (dashed line) approaches $1/\sqrt{2\pi}$ asymptotically. All curves for which $\lambda > 4$ have approximately the shape of the curve plotted with $\lambda = 4$, their peaks coming at points farther out. For any λ , the peak occurs at approximately $\sqrt{1+2\lambda}$.

The function of a so-called "linear detector" is to measure or "trace" an envelope. (There would be less confusion among radio engineers if the device were simply named definitively, "envelope tracer.") Circuit design to minimize tracking failures (distortion) is so well understood and generally so economically achieved that the envelope tracer is now almost universally chosen in preference to any other type of detector. The curves of Fig. 1 illustrate the output of a receiver using an envelope tracer and Type-A scan. The x axis corresponds to the vertical distance along a "picture element." The value of $P(x)$ illustrates the average in-

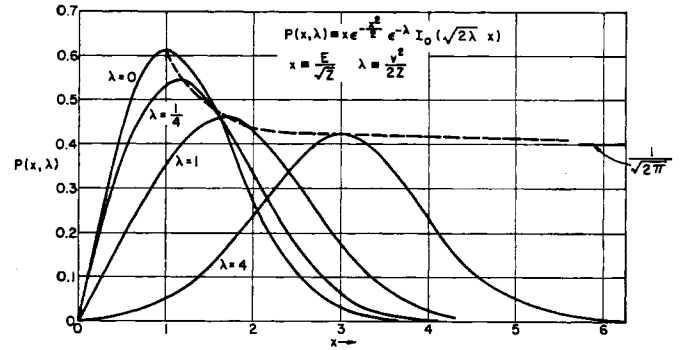


Fig. 1—Density of probability for the envelope of the sum of a sine wave and normal noise.

tensity of illumination at the corresponding height. One recognizes certain familiar phenomena, the light-intensity distribution for noise alone, the darkening of the base and particularly the identifying increase in contrast in the vicinity of $2 < x < 3$ as the echo (v) is brought up from zero to a barely discernible value.

The expectation of E , that is, the average value of the envelope of a mixture of sine signal and noise, is

$$\begin{aligned} \bar{E} &= \sqrt{\frac{\pi Z}{2}} e^{-\lambda/2} [(1+\lambda)I_0(\lambda/2) + \lambda I_1(\lambda/2)] \\ &= \sqrt{\frac{\pi Z}{2}} [1 + \lambda + O(\lambda^2)] \\ &\sim \sqrt{v^2 + Z + O(\lambda^{-2})}. \end{aligned} \quad (14)$$

On the basis of (14), diode envelope tracers metered in the condenser leak have for several years supplanted thermocouples in some of our laboratory equipment for measuring receiver noise factor and fundamental fluctuation effects in terms of the output of a calibrated signal generator. Far more rugged than thermocouples, they have proved extremely reliable as well.

III. INTEGRATION BEFORE DETECTION

The possibility of integration before detection is particularly attractive, as compared with postdetector integration, because it affords an opportunity to add directly the separate responses of the receiver to a series of successive echoes from the target, whereas a similar aggregate of the unavoidable noise pulses tends toward an rms sum, the phases being at random. Such a purpose can be achieved only when one has fulfilled an all important condition: the RF phase of the echoes must vary uniformly with time throughout the integrating period. This implies at least three subsidiary conditions.

- a) The target must not move along the range an appreciable fraction of a wavelength during the integrating period. A shift of $\frac{1}{4}$ wavelength reverses the echo phase.

¹ The symbol $O(\lambda^2)$ in (13a) is read "order of λ^2 ."

- b) The transmitter must be pulsed in such a way as not to disturb the phase continuity. Conventional pulsing methods fail in this regard.
- c) The combination of transmitter and local oscillator must produce a stable frequency in the receiver IF channel.

Means for satisfying b) and c) are at hand. But a) appears to confine this method, in its simplest form, to relatively long-wave systems, for, although one can devise means for accommodating a target moving with a known speed, obviously no immutable assumption as to speed can be made while the target is, to the observer, only a suspicion.

Assuming that these conditions can be met, we picture a suitable scheme as follows: Between the antenna and a "gate" is provided a bandwidth (RF and IF) at least sufficient to pass the rectangular pulse (1) without appreciable shape distortion; an even larger bandwidth will do no harm. The gate energizes the succeeding equipment periodically for an interval (τ) just sufficient to pass an echo from the segment of range under examination, and thus performs the important function of excluding from the integrator all echoes from other parts of the range and all noise save that which unavoidably passes through the gate.

The signal from a fixed target comes through as a periodic modulation of the carrier. And a Fourier analysis portrays the signal as a series of discrete continuous waves symmetrically disposed around the carrier frequency with a uniform frequency spacing of $1/T$ and amplitudes depending, in general, upon the nature of the pulsing function. A primitive integrator consists of a channel whose bandwidth is small relative to $1/T$ and centered on one of the components, excluding the others. The smaller the bandwidth, the better the "memory," i.e., the longer the effective integration time. A more complex integrator would make use of a larger fraction of the available signal power with a battery of primitive integrators individually centered on a large number of wave components, their outputs combined before detection. An analysis of the primitive integrator will suffice for the present.

Consider the rectangular pulsing function (2). Let the narrow filter have a noise bandwidth Δf , and be centered on the carrier frequency. At the antenna terminals the only Fourier component of signal able to pass the filter has an amplitude $V(\tau/T)$. The equivalent mean-square noise voltage at the same point is $a\Delta f$, of which the fraction passing the gate is $a\Delta f(\tau/T)$. Since integration is negligible unless $\Delta f \ll 1/T$, it will be correct to assume that the filter output shows no trace of gating modulation of the noise. The envelope probability function for the filter output (13) is therefore characterized by

$$\lambda = \frac{1}{2} \left(\frac{V\tau}{T} \right)^2 \left(\frac{T}{a\Delta f\tau} \right) = H\lambda_0, \quad (15)$$

where λ_0 is defined by (6) and

$$H = \frac{1}{T\Delta f},$$

shown later to be the effective number of pulses memorized by the integrator.

For illustration we suppose the filter output put through a metered envelope tracer.² For simplicity, consider the meter to have zero time constant; clearly a time constant in excess of $1/\Delta f$ implies further integration in the meter itself. In a similar way, observer integration, with lowering of minimum visible signal, is the obvious consequence of a prolonged observation of the meter. Both of these unnecessarily complicate the picture. We shall assume simply that a meter reading in excess of a certain bias causes a gun to be fired. We shall also assume that the minimum visible signal corresponds to a value of λ large enough that (13b) can be used.

We can afford to be neither unduly spendthrift with ammunition nor overly frugal. We specify a fraction of the time r_0 in which we will permit errors (firing of gun) in the absence of a target; and we define the minimum visible signal to correspond with the value of λ for which, in the presence of a target, errors occur (fire withheld) a fraction of the time r . That is, the bias b is chosen so that in the presence of noise alone, the chance that the output voltage exceeds the bias is r_0 ; the minimum visible signal is then that for which the chance is r that, in the presence of the target, the output voltage falls below the bias. Thus

$$r_0 = \int_b^\infty P(x, 0) dx = e^{-b^2/2}$$

$$r = \int_0^b P(x, \lambda) dx \approx \frac{1}{2} \left[1 - \phi \left(\frac{\sqrt{1+2\lambda} - b}{\sqrt{2}} \right) \right],$$

where

$$\phi(x) \equiv \text{erf } x \equiv \frac{2}{\sqrt{\pi}} \int_0^x e^{-y^2} dy \quad (16)$$

yielding

$$b = \sqrt{2 \log (1/r_0)}$$

$$\sqrt{1+2H\lambda_0} \equiv \sqrt{1+2\lambda} = \sqrt{2 \log (1/r_0)} + S(r) \quad (17)$$

where

$$S(x) \equiv \sqrt{2} \phi^{-1}(1-2x),$$

² A system proposed by T. T. Eaton dispenses with this detector by operating the local oscillator at carrier frequency, essentially the same method but with a zero-frequency IF. The chance of a local-oscillator phase such as to produce FM without AM could be eliminated by filtering off half the echo sidebands, yet the fundamental difficulty with moving targets remains and would be evidenced by a change in sign of the amplitude modulation.

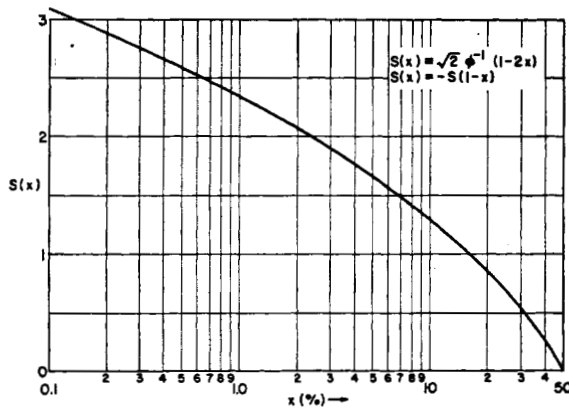


Fig. 2—Plot of $S(x)$ vs x expressed as per cent.

$\phi^{-1}(x)$ being the inverse (not the reciprocal) of the error function. $S(x)$ is graphed on Fig. 2.

To illustrate, suppose $r_0 = 0.05$, $r = 0.20$, $H = 10^4$. Then $b = 2.45$, and since it is measured on the noise scale ($\bar{x} = \sqrt{\pi}/2$, $\lambda = 0$), the result means that the bias is established at 1.96 times the long-time-average noise output. We find, also, $\lambda = 4.8$, justifying the use of (13b) for this example. In the presence of this signal, the long-time-average output is 2.62 times the long-time-average noise output. Finally,

$$\lambda_0 = \lambda/H = 4.8 \times 10^{-4}. \quad (18)$$

It might be helpful to express this result in terms of a signal-to-noise ratio. It will be shown later that the optimum predetector noise bandwidth for a system using *postdetection* integration in any amount (such as the conventional Type-A scan) is $\Delta f = 1/\tau$ when, as in this example, the pulsing function is (2). A system using this predetector bandwidth will be called a "standard system" for reference purposes, so as to define a standard of noise bandwidth. The signal-to-noise ratio (S/N) will then be defined as the ratio of the rms value of the peak signal to the rms value of equivalent noise voltage at the antenna terminals of a *standard* system. Reference to (6) shows that, for rectangular pulses,

$$S/N = \lambda_0^{1/2}. \quad (19)$$

A standard of this sort will appear less arbitrary when it is appreciated that the minimum visible signal in a standard system using Type-A scan is generally observed to have a magnitude such that

$$S/N = \lambda_0^{1/2} = O(1). \quad (20)$$

From (18) we have

$$S/N = \lambda_0^{1/2} = 0.022, \quad (21)$$

an indicated improvement of some 46-fold in minimum visible signal. In a system using a pulse repetition rate of 1000 sec^{-1} , a filter with $\Delta f = 0.1 \text{ cps}$ would be required.

An important part of this analysis is to determine

what is wanted of the transmitter. That is, what criterion of transmitter operation measures its contribution to S/N for a target of fixed size and range? From (17) it is clear that, for a receiver with a fixed predetector integration time, the criterion is long-time-average power. With the transmitter producing maximum average power, pulse length and repetition rate do not otherwise affect S/N , and may be adjusted to meet other operating demands, such as range or range resolution.

An approach to the composite integrator mentioned earlier is best understood by considering at first just two integrating channels, one centered on the carrier frequency, the other spaced from it by a frequency interval $1/T$, both possessing the same noise bandwidth, hence the same H . An analysis shows that, because the frequency spacing is an integral multiple of the gating frequency, the noise output from the additional channel is *not* randomly related to the output from the first channel. On the contrary, except for the small shift in mean frequency, there is virtually perfect instantaneous correlation of the two envelopes, provided, of course, that the time delay is the same through both channels. It follows that a composite integrator cannot yield a better S/N than may be obtained with a single channel.

However, the composite integrator could, in theory, be utilized to side-step the necessity for gating. Suppose the number of channels M increased to the point where $M \approx 2T/\tau$, beyond which the Fourier components of the echo fall off too rapidly to be useful. If, in addition the time delay were made uniform over all channels, and the outputs combined before detection, no gate would be necessary, for a composite of H pulses would automatically be constructed, and the whole range could be thrown on, say, a Type-A scope. A broad-band regenerative loop with time delay equal to the pulsing period would theoretically fulfill this purpose.³

But, aside from practical constructional difficulties, the outlook appears discouraging from the point of view of targets in motion, or any other disruption of carrier-phase continuity. To illustrate, consider the primitive integrator analyzed. If the carrier phase jumps at random between echoes, nothing is left to distinguish signal from noise, save amplitude. All the benefits of integration are lost. The performance is then that formulated by (17) with $H=1$, yielding, in the numerical illustration chosen,

$$S/N = \sqrt{4.8} = 2.2. \quad (22)$$

IV. INTEGRATION AFTER DETECTION

Continuity of carrier phase is here of no concern. A typical arrangement follows a gate with a detector and an integrator in sequence. To begin with, we shall

³ In this connection see [5], Section 12.

assume a stationary target and an integrator with perfect memory (e.g., a counter or a fluxmeter) capable of integrating any desired number of echoes N , generalizing afterwards to integrators with imperfect memories and to targets in motion.

It is the detector's function to measure the envelope of each pulse as it comes through the gate. At some point in each pulse the echo power (if any) will reach its maximum value, whereas the noise power is distributed at random in time. Clearly, therefore, the maximum reporting accuracy is achieved with the use of a properly centered gate having an *infinitesimal* width.

A. Predetector Selectivity

In these circumstances reporting accuracy is best served by arranging the selectivity so that the peak signal presented to the detector bears as high a ratio as possible to the rms noise amplitude at the gate. That is, given a value of λ_0 at the antenna terminals (6), what transfer characteristic gives maximum λ at the detector?

Consider a *single* echo of the form (1). It has a Laplace transform

$$E(p) = \int_{-\infty}^{\infty} e(t) e^{-pt} dt$$

from which the mate transformation permits recovery of the original, viz.,

$$e(t) = \frac{1}{2\pi i} \oint E(p) e^{pt} dp.$$

Let $G(p)$ denote the transfer characteristic between antenna and detector. The voltage at the detector input will be

$$V(t) = \frac{1}{2\pi i} \oint E(p) G(p) e^{pt} dp.$$

At some instant t' , depending, *inter alia*, upon the time delay and the nature of $e(t)$, this voltage will be a maximum $V(t')$.

The mean-square noise at the gate is $(G_0^2 a \Delta f)$, where G_0 is the absolute value of the gain at some reference frequency (for convenience, the carrier frequency) and the noise bandwidth is, as usual, given by

$$\Delta f \equiv \frac{1}{G_0^2} \int_0^{\infty} |G|^2 df = \frac{1}{4\pi i G_0^2} \oint G(p) G(-p) dp. \quad (23)$$

From the definition (12) we find, at the detector input,

$$\lambda = \frac{V^2(t')}{2G_0^2 a \Delta f} = \frac{W}{a} \cdot \frac{V^2(t')}{2WG_0^2 \Delta f} = \lambda_0 \cdot U \quad (24)$$

where

$$U \equiv \frac{V^2(t')}{2WG_0^2 \Delta f}. \quad (24a)$$

Our problem is to find, while holding G_0 fixed, the form of $G(p)$ which makes U a maximum. The logarithmic variational of (24a) leads to

$$\frac{2\delta[V(t')]}{V(t')} - \frac{\delta[\Delta f]}{\Delta f} = 0$$

or

$$\oint \delta G(p) \left\{ \frac{E(p) e^{pt'}}{V(t')} - \frac{G(-p)}{2G_0^2 \Delta f} \right\} dp = 0,$$

whence, if $E_0 \equiv |E(p)|$ for the value of p which makes $|G(p)| = G_0$, say $p = p_0 = i\omega_0$, then

$$\frac{G(p)}{G(p_0)} = \frac{E(-p)}{E(-p_0)} e^{-(p-p_0)t'} \quad (25)$$

and

$$V(t') = 2G_0 E_0 \Delta f. \quad (26)$$

This says that, except for the necessary time-shift factor, $e^{-pt'}$, the best transfer characteristic looks like the conjugate of the spectrum of the echo at the antenna. The existence of such a transfer characteristic for a finite, lumped-element transducer is in doubt, but we can certainly approximate (25) while expecting to approximate its benefits.⁴

With (25) we find, from (23),

$$\Delta f = \frac{W}{2E_0^2}. \quad (27)$$

With this and (26) we find, from (24a), that the maximum value of U is

$$U = 1. \quad (28)$$

Consider now a specific example, a carrier pulsed with the rectangular pulsing function (2); we have

$$W = \frac{1}{2} V^2 \tau$$

$$E(p) = V \left[\frac{\sinh(p - p_0) \frac{\tau}{2}}{p - p_0} + \frac{\sinh(p + p_0) \frac{\tau}{2}}{p + p_0} \right].$$

Notice that $E(-p) = E(p)$, that it has no poles in the finite domain, and can therefore be only approximated by means of a physical transducer. Also notice that, if $\omega_0 \gg 2/\tau$,

$$E_0 = \frac{V\tau}{2},$$

⁴ In a Radiation Laboratory report [4], H. M. James delineates a theory of N. Wiener's leading to $G(p) \propto |E(p)|^2$, but his basic criterion of "goodness" is not germane to the problem in hand.

whence (27) becomes $\Delta f = 1/\tau$ and (25) becomes

$$\frac{G(p)}{G_0} = \frac{E(-p)}{E_0} = \frac{\sin x}{x} e^{-(i2t'/\tau)x} \quad (29)$$

where

$$x \equiv (\omega - \omega_0) \frac{\tau}{2}.$$

The actual shape of the signal pulse at the gate $V(t)$ is an isosceles triangle with a base, 2τ , twice the length of the rectangular pulse supplied the transducer.

In attempting to approximate (29) with a physical network, it is probably sufficient to carry the likeness out to $x = \pm\pi$ only. To throw away the rest introduces rather negligible "tears" on the output pulse and hardly impairs the value of U and λ . In any event, it is clear that the *shape* of the pulse is never a factor in the recognition of small signals.

While it is valuable to know what arrangement yields maximum U for a rectangular pulse, it is of even more interest to know how rapidly U falls off if one adopts certain engineering approximations to the ideal. Two cases will be discussed.

First, consider a uniform, rectangular characteristic with linear phase shift (still a characteristic realized physically only in approximation), *i.e.*,

$$G(p) = G_0 e^{-(p-p_0)t'}, \quad -\frac{\alpha\pi}{2} \leq x \leq \frac{\alpha\pi}{2}, \quad x \equiv (\omega - \omega_0) \frac{\tau}{2}.$$

For this filter the output wave shape is the familiar integral-sine function. And we find

$$V(t') = G_0 V \frac{2}{\pi} Si\left(\frac{\alpha\pi}{2}\right); \quad Si(x) \equiv \int_0^x \frac{\sin y}{y} dy$$

$$\Delta f = \frac{\alpha}{\tau}.$$

That is, when $\alpha=1$, the noise bandwidth equals the standard noise bandwidth $1/\tau$. From (24a) there results

$$U = U_1 \equiv \frac{4 Si^2\left(\frac{\alpha\pi}{2}\right)}{\pi^2 \alpha}. \quad (30)$$

Second, consider a cascade of n identical stages, each consisting of a high- Q circuit of one degree of freedom. Define γ to be the nominal (3 db down) bandwidth in radian measure of one such stage. The nominal bandwidth of the whole transducer is

$$\gamma_n = \gamma \sqrt{2^{1/n} - 1} \sim \gamma \sqrt{\frac{\log 2}{n}}.$$

The output pulse is

$$V(t) \approx G_0 V \cos \omega_0 t \cdot \frac{1}{2} \left\{ \phi \left[\frac{\gamma}{2\sqrt{2(n-1)}} \left(t - t' + \frac{\tau}{2} \right) \right] - \phi \left[\frac{\gamma}{2\sqrt{2(n-1)}} \left(t - t' - \frac{\tau}{2} \right) \right] \right\},$$

where ϕ is the error function of (16), and

$$t' \equiv \frac{(2n-1)}{\gamma} = \text{the delay time.}$$

We find, therefore

$$V(t') = G_0 V \phi \left[\frac{\gamma\tau}{4\sqrt{2(n-1)}} \right] \sim G_0 V \phi \left[\frac{\gamma_n\tau}{4\sqrt{2 \log 2}} \right].$$

We also find from (23)

$$\Delta f = \frac{[2(n-1)]!}{2^{2n}[(n-1)!]^2} \gamma \sim \frac{\gamma_n}{4\sqrt{\pi \log 2}},$$

which shows that for large n the noise bandwidth is 6 per cent greater than the nominal bandwidth (cps).

As before, define α with

$$\Delta f \equiv \frac{\alpha}{\tau},$$

so that when $\alpha=1$, the noise bandwidth equals the standard noise bandwidth $1/\tau$. From (24a) there results

$$U = U_2 \equiv \frac{1}{\alpha} \phi^2 \left(\alpha \sqrt{\frac{\pi}{2}} \right). \quad (31)$$

U_1 and U_2 are graphed in Fig. 3, which shows for the first transducer an optimum (noise) bandwidth of about 4/3 the standard noise bandwidth, for the second an optimum noise bandwidth of about 4/5 the standard, and indicates generally that a satisfactory approach to the maximum, $U=1$, is not critical and can be achieved readily enough by normal engineering methods.

No matter which transducer is used, it is impossible at present to discuss with precision how reporting accuracy is impaired (U decreased, in effect) by a widening of the gate from the infinitesimal width assumed above. To do so requires the development of a correlation function which relates the probability function (8) for the instantaneous value of noise at a given instant to its *observed* value at a prior instant, *i.e.*, a development which injects the frequency spectrum into the statistics. However, if the gate were widened, perhaps to accommodate a moving target, no major change in the effective value of U would occur until the gate passed most of the pulse $V(t)$, for the rate at which the noise envelope may change is controlled by Δf . Further

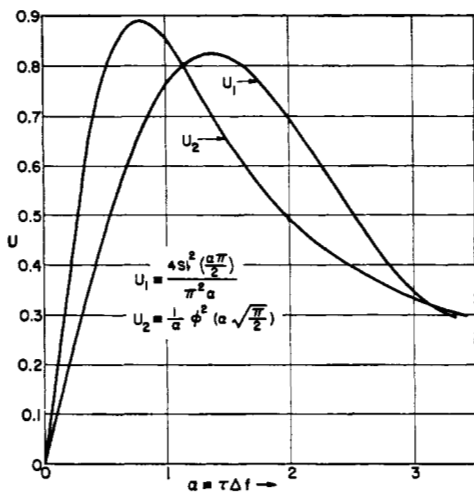


Fig. 3—Two engineering approximations to the optimum filter for rectangular pulses. U_1 refers to a rectangular transfer characteristic, with linear phase shift. U_2 refers to a cascade of n identical simple-tuned stages.

widening of the gate by a factor m would alter effective U to approximately $U/\sqrt{m}$. One could, alternatively, keep the gate infinitesimal and narrow Δf so as artificially to increase the pulse length at the detector. The more or less straight slope to the left-hand side of the curves of Fig. 3 indicates that if, at the start of this maneuver, Δf had been optimum for a fixed target, a reduction of Δf by a factor m would provide the desired accommodation for a moving target, but would alter effective U to about U/m . In comparison with widening the gate, this practice is seen to be inferior. The best type of accommodation appears to involve holding the gate infinitesimal while increasing the length of the transmitted pulse and, of course, adjusting Δf to suit, a process which leaves U unaffected. If, in this procedure, the pulse energy (W) is unaltered, there is no change whatsoever in the value of λ . That is, the moving target is detected with the same accuracy as, formerly, the stationary target. The only sacrifice is range resolution, a property of generally minor importance at the range limit. One may even stand to gain accuracy in this procedure, for, as we shall see later, with a longer pulse it will often be possible to improve the transmitter criterion.

B. The Statistical Problem

When the target is stationary and the gate infinitesimal, N successive echoes reach the detector, all characterized by the same λ , but each possessing independently an envelope described by the density-of-probability function $P(x, \lambda)$ of (13). Each, in turn, goes through the detector, which measures its envelope, and then into the integrator where the sum of N such detector operations is the final step. An envelope tracer (unbiased linear detector) reports the height (x) of the envelope; a so-called square-law detector (unbiased) reports x^2 to

the integrator; and, in general, a detector of arbitrary characteristic may be said to report $f(x)$.

The report of the integrator, namely,

$$y \equiv \sum_1^N f(x_k) \quad (32)$$

can now be calculated by a direct application of the Laplace-Liapounoff theorem (see Section II-A), each $f(x)$ being regarded as an independent chance variable in the sense of that theorem. Let

$$\bar{f} \equiv \bar{f}(\lambda) \equiv \int_0^\infty f(x) P(x, \lambda) dx; \quad \bar{f}_0 \equiv \bar{f}(0). \quad (33)$$

We have

$$\bar{y} = N\bar{f}; \quad \bar{y}_0 = N\bar{f}_0$$

$$B = N(\bar{f}^2 - \bar{f}_0^2); \quad B_0 = N(\bar{f}_0^2 - \bar{f}_0^2).$$

Then, preserving our former definitions of integrator bias b , and permissible errors r_0 and r , we have as before two expressions which define the bias and the minimum visible value of λ ,

$$r_0 = \frac{1}{2} \left[1 - \phi \left(\frac{b - \bar{y}_0}{\sqrt{2B_0}} \right) \right]$$

$$r = \frac{1}{2} \left[1 - \phi \left(\frac{\bar{y} - b}{\sqrt{2B}} \right) \right],$$

which upon inversion yield

$$\left. \begin{aligned} b - \bar{y}_0 &= B_0^{1/2} S(r_0) \\ \bar{y} - b &= B^{1/2} S(r) \end{aligned} \right\} \quad (34)$$

where $S(x) \equiv \sqrt{2}\phi^{-1}(1-2x)$, graphed in Fig. 2.

The first of these equations defines b . In taking the sum, to formulate λ , we are generally entitled to assume $\lambda \ll 1$, expanding $\bar{y}$ as follows:

$$\bar{y} \approx \bar{y}_0 + N\lambda \left(\frac{\partial \bar{f}}{\partial \lambda} \right)_0, \quad (35)$$

and to ignore the difference between B and B_0 . This yields

$$\lambda = \sqrt{\frac{D}{N}} [S(r_0) + S(r)] \quad (36)$$

where

$$D \equiv \frac{\bar{f}_0^2 - \bar{f}_0^2}{\left(\frac{\partial \bar{f}}{\partial \lambda} \right)_0^2}, \quad (37)$$

a property of the detector alone.

Bringing in the predetector selectivity, we have finally the minimum visible λ_0

$$\lambda_0 = \frac{1}{U} \sqrt{\frac{D}{N}} [S(r_0) + S(r)]. \quad (38)$$

C. Transmitter Criterion

In contrast with the earlier observation, in respect to integration before detection, that maximum *average* power was wanted from the transmitter, we note in this case that the criterion is maximum

$$\text{Pulse energy} \times \text{Average power} \quad (39)$$

Presumably the average power of a transmitting tube is set by its dissipation limit. Therefore (39) calls for its operation at that limit, and at the lowest possible pulse rate. Where maximum pulse length is set by a specified range resolution or by a specified minimum working range, the lowest possible pulse rate will then be fixed at the point where the tube fails to meet the increasing demand for peak power. For long-range detection work, where range resolution is not so important, there is therefore a distinct advantage in the use of long pulses, particularly since, as we have seen, it accords with the best method for accommodating moving targets.

D. Range vs Integration Time

If one assumes inverse-square diffusion of radiated power, the echo energy is inversely proportional to the fourth power of the range

$$W \propto r^{-4}$$

for a target moving outward along the range. In order that the target be just discernible at each point of the range, the number of echoes integrated must increase

$$N \propto r^3.$$

This accounts for the extremely sharp range limit often observed on systems in which the integration time is fixed.

Where the attenuation increases, beyond line-of-sight, for example, the exponent of r increases.

E. Comparison of Detectors

The detector function (37) is analyzed with the aid of

$$\left. \begin{aligned} \bar{f}_0 &= \int_0^\infty f(x) x e^{-x^2/2} dx \\ \bar{f}_0^2 &= \int_0^\infty f^2(x) x e^{-x^2/2} dx \\ \left(\frac{\partial \bar{f}}{\partial \lambda} \right)_0 &= \frac{1}{2} \int_0^\infty f(x) x (x^2 - 2) e^{-x^2/2} dx \end{aligned} \right\} \quad (40)$$

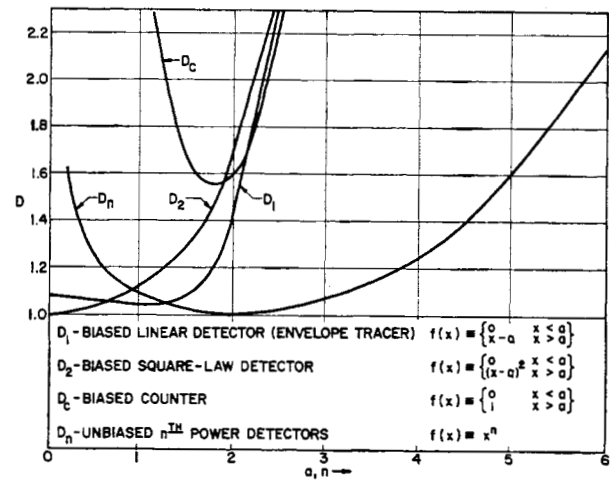


Fig. 4—The detector function D for a variety of detector types.

An application of the variational calculus shows that D has a minimum value

$$D = 1 \quad (41)$$

when

$$f(x) = A + Bx^2, \quad A \text{ and } B \text{ arbitrary.}$$

That is to say, of all detectors, the unbiased square-law detector is the best possible for this purpose. The *biased* square-law detector is not a member of this group. However, some other common detectors perform in an adequate fashion.⁵ On Fig. 4 are graphed the values of D for the following:

D_1 : Biased linear detector (envelope tracer);

$$f(x) = \begin{cases} 0 & x < a \\ x-a & x > a \end{cases};$$

D_2 : Biased square-law detector; $f(x) = \begin{cases} 0 & x < a \\ (x-a)^2 & x > a \end{cases};$

D_c : Biased counter; $f(x) = \begin{cases} 0 & x < a \\ 1 & x > a \end{cases};$

D_n : Unbiased n^{th} power detectors $f(x) = x^n$.

The biased counter, for example, works best with a bias equal to 1.8 times the rms noise level, $a = 1.8$, but is hardly in the running.

F. Minimum Visible Signal

To illustrate the results achieved with this kind of integration, let us use in (38), $U = 1$, $D = 1$, and, as before, $r_0 = 0.05$, $r = 0.20$, $N = 10^4$. With the definition (19) we have

$$S/N = \lambda_0^{1/2} = 0.16,$$

cf., (20), (21), and (22). In a system using a pulse repetition rate of 1000 sec⁻¹, an integration time of 10 sec would be required.

⁵ See [5], Section 4.

G. Integrators with Imperfect Memory

Heretofore a perfect memory has been assumed. Let us now examine the effects of a faulty memory, but one good enough to remember a large number of pulses, that is, a large enough decay time (T_0) that $T_0 \gg T$. A typical primitive integrator of this kind consists of a narrow filter centered on a harmonic of the pulse frequency $1/T$. We introduce a function A_k to describe the memory of the integrator, and, in place of (32), write

$$y = \sum_1^N f(x_k) A_k,$$

where $f(x_k)$ is the k th pulse supplied by the detector and $f(x_k)A_k$ is the recollection of that pulse by the integrator after N such pulses. The report y of such an integrator will be nearly sinusoidal, at the filter frequency, and may be read at any instant of the interval between the N th and $(N+1)$ st pulses with uniform accuracy. In practice one would therefore feed the filter output to an envelope tracer (linear detector), using its output as a report. By the same argument, the analysis will not suffer if we write

$$y = \sum_0^{N-1} f(x_k) A(kT), \quad (42)$$

where $A(t)$ is the response of the integrator to unit impulse applied at $t=0$. The implication is

$$Z(p) = \int_0^\infty A(t) e^{-pt} dt, \quad A(t) = \frac{1}{2\pi i} \oint Z(p) e^{pt} dp, \quad (43)$$

where $Z(p)$ is the complex transfer characteristic of the filter. This approach leads to (38) with N replaced by

$$H(N) = \frac{\left[\sum_0^{N-1} A(kT) \right]^2}{\sum_0^{N-1} A^2(kT)}. \quad (44)$$

One sees that the maximum value of H accompanies a perfect memory, *i.e.*, $A = \text{constant}$, in which case $H = N$. For any system with imperfect memory, H is also a function of N , but becomes virtually independent of N when the storage time exceeds the nominal memory time; this value of H , namely $H(\infty)$, is of most interest.

By interpreting (44) in the light of (43) one finds, after some labor, the following summarized results:

$$\left. \begin{aligned} H(N) &= N: && \text{perfect memory} \\ H(\infty) &= \frac{1}{2T\Delta f}: && \text{primitive aperiodic integrator} \\ H(\infty) &= \frac{1}{T\Delta f}: && \text{primitive harmonic integrator} \end{aligned} \right\} \quad (45)$$

where Δf is the noise bandwidth of the filter.

So far as S/N is concerned there is nothing here to choose between an aperiodic integrator and a harmonic integrator working at one of the multiples of the repetition frequency. But the outstanding difficulties with harmonic integrators are the necessity for retuning with a change in pulse rate, and the faults that occur if the pulsing frequency wobbles sufficiently to spill over the edge of the filter. The aperiodic integrator suffers in neither of these ways, and long time constants are achieved without difficulty: 1 microfarad + 10 megohms $\rightarrow$ 10 sec.

The following examples illustrate the way in which H builds up during an "exposure," and show the magnitudes of and connection between Δf and T_0 for two kinds of primitive integrator.

Consider an aperiodic integrator consisting of R , C in shunt. Defining $T_0 \equiv RC$, we have

$$\Delta f = \frac{1}{4RC} = \frac{1}{4T_0}$$

$$H(N) = \frac{\frac{\tanh \frac{NT}{2T_0}}{2T_0}}{\frac{T}{\tanh \frac{T}{2T_0}}} \xrightarrow{N \rightarrow \infty} \coth \frac{T}{2T_0} \approx \frac{2T_0}{T}.$$

If this is changed to a harmonic integrator by adding shunt inductance to resonate at a harmonic of the pulse frequency, Δf is unchanged,

$$H(N) = \frac{\frac{\tanh \frac{NT}{4T_0}}{4T_0}}{\frac{T}{\tanh \frac{T}{4T_0}}} \xrightarrow{N \rightarrow \infty} \coth \frac{T}{4T_0} \approx \frac{4T_0}{T},$$

twice the aperiodic figure, because the latter filter has one half the decrement of the other.

The observed failure of long-persistence fluorescent materials to improve S/N in conventional visual systems,⁶ even in the case of phosphors which show a visible residual days after excitation, is probably traceable to the fact that although the decay time may, by arbitrary definition, be said to exceed notably the 1/15-sec integrating period of the eye, nevertheless, insofar as these phosphors can be considered linear systems, the noise bandwidth

$$\Delta f = \frac{1}{2} \frac{\int_0^\infty A^2(t) dt}{\left[\int_0^\infty A(t) dt \right]^2}$$

is too great to be helpful, *i.e.*, $\Delta f > 15$ cps.

⁶ See [5], Section 7.

H. Composite Integrators

Finally one asks whether S/N may be improved by combining the reports of several integrating systems. If one combines the outputs of two identical adjacent receivers, each complete with its own antenna, the answer is "yes" if the noise sources are all within the receivers, probably "no" if they are outside. A gain of this kind is only what one might secure with a single system and an antenna of twice the directivity [1].

To realize that the reporting accuracy of a system consisting of a battery of primitive harmonic integrators fed from one detector is better than that of the poorest, not better than that of the best, hardly requires the formal proof available. For each primitive integrator is fed the same information as the others. Consequently, while the use of a broad-band time-delay regenerative loop *after* detection provides, as before, a convenient means for throwing the whole range at once on a screen, the minimum visible signal is still not better than could be achieved with a primitive integrator having the same (per capita) noise bandwidth.⁷

V. VISUAL AND PHOTOGRAPHIC METHODS

Since this subject comes face-to-face with problems of observer behavior, the theories propounded and the results formulated are claimed to be but rudimentary approaches to the matter. For this reason only rough estimates will be made, and certain refinements ignored.

It will be assumed that the integrating period of the eye is about 1/15 sec; for want of information, that the stimulus of N pulses arriving during an integrating period is independent of the distribution of the pulses over the period; and that the eye distinguishes signal from noise *only* on the basis of light contrasts.

A. Brightness Modulation

In this scheme, the detector output is applied to the grid of a cathode-ray tube, a sweep voltage being applied to the horizontal plates, and, in its simplest form, to the vertical plates, nothing. It is assumed that the eye contrasts the brightness (y) where an echo is sought with the average brightness ($\bar{y}_0$) where there is known to be noise only. (Actually, I suppose the comparison is not with $\bar{y}_0$ but with the simultaneous brightness observed in, perhaps, an adjacent "picture element," but the result will be much the same.) For convenience, let us assume a standard predetector bandwidth, $U=1$; an infinitesimal gate, provided by the eye; and, including the conversion into light, a detection process represented by an unbiased linear detector, $D=1.08 \approx 1$.

Now, as before, we suppose identification of a target satisfactory only when y lies above a certain bias for all but a fraction r of the time. And here the Weber-Fechner law enters to say that a brightness less than $(1+\epsilon)\bar{y}_0$ cannot be discerned, where, under ideal con-

ditions [6], ϵ may lie in the range $0.02 < \epsilon < 0.04$, but may conceivably be considerably greater in the case of transient stimuli. From (34) this leads to

$$\lambda_0 \approx 2\epsilon + \frac{S(r)}{\sqrt{N}}. \quad (46)$$

To illustrate, suppose $\epsilon=0.04$, $r=0.05$, and compare the minimum visible signals for two pulse rates, 60 and 600 cps. We find

$$(S/N)_{60} \equiv \lambda_0^{1/2} = 0.95$$

$$(S/N)_{600} \equiv \lambda_0^{1/2} = 0.58.$$

Their ratio is 1.6. In the report PTR-7C [5], the minimum visible signal was observed to drop by a factor of 1.7 in the switch from 60 to 600 cps.⁸

The presence of the ϵ term vitiates the relation

$$(S/N) \propto N^{-1/4}$$

eventually in experiments of this sort in which the pulse rate is raised ad infinitum. For if $\epsilon=0.04$, then

$$(S/N)_\infty = \sqrt{2\epsilon} = 0.28,$$

this limit being established solely by the inability of the eye to discern weak contrasts, a limitation to which electromechanical indicating systems which produce algebraic rather than logarithmic reports are not subject. The report PTR-7C, for example, describes tests⁹ with a primitive aperiodic integrator for which the minimum visible signal lay below that of a Type-A scan by a factor of 10.

A similar limitation determines the ultimate effectiveness of film used, let us say, to increase the integration period beyond 1/15 sec. Expression (46) is still applicable, with N representing the number of echoes in an exposure. But ϵ is not reduced unless the gamma of the photographic process exceeds unity. In one test,¹⁰ a 2-volt signal was barely visible on the screen, stood out clearly on a 5-sec film exposure; yet a $\frac{1}{2}$ -volt pulse was not visible on a 5-sec exposure.

B. Deflection Modulation (Type-A Scan)

In this scheme the signal is used, not to modulate the intensity of the beam, but to deflect it vertically. It will be assumed here that a signal is recognized in terms of the light contrast produced at a certain altitude. We have seen, Fig. 4, that the biased counter rapidly becomes inefficient as the bias is raised above its optimum value; similarly, visual tests show that the reporting accuracy of an observer whose vision is confined to the higher altitudes is inferior. The range of altitudes in

⁷ See [5], Section 12.

⁸ See [5], Section 5.

⁹ See [5], Section 9.

¹⁰ See [5], Section 8.

which the contrast is sought will therefore be assumed small and situated in the vicinity of $x=1$.

With the same assumptions as before, *i.e.*, standard bandwidth, linear detector, etc., we start with the observation that the probability of altitude, x , $x+\Delta x$ on a given sweep is, from (13), $P(x, \lambda)\Delta x$. After some labor, and the usual simplifying assumptions (large N , small λ , etc.) we establish the probability that the light output at the altitude x in the "picture element" where signal is sought should exceed by the fraction ϵ for all but a fraction of the time r , the average light output from a similar region exposed to noise alone.

This leads to

$$\lambda_0 \approx \frac{2\epsilon + \frac{2}{\sqrt{N}} \sqrt{\frac{e^{x^2/2}}{x\Delta x}} S(r)}{x^2 - 2} \quad (47)$$

Presumably the eye seeks evidence over a considerable range of altitudes, and a correct formulation would rid (47) of Δx by some kind of altitude integration. For this crude attack it will suffice to write $\Delta x=1$. That reporting accuracy is nil in the vicinity of $x=\sqrt{2}$ is illustrated by (13) and the curves of Fig. 1. Ignoring ϵ , minimum λ_0 occurs in the vicinity of $x=2.6$, which is about the height at which signal is actually first seen. Using these values of x , Δx , we have

$$\lambda_0 \approx 0.4\epsilon + \sqrt{\frac{2}{N}} S(r). \quad (48)$$

A comparison of (46) and (48) attests and helps to explain the experimental conclusion¹¹ that minimum visible signal is about the same with either type of modulation.

Just as film may be used to increase the integrating period beyond 1/15 sec, so will other devices. The insertion of a time-delay regenerative loop between detector and cathode-ray tube could serve this purpose with either type of modulation, but the minimum visible signal would still be limited by ϵ . Even should the loop decay time be well below 1/15 sec, its insertion would reduce the minimum signal visible with deflection modulation.¹² For the effect of the insertion would be to form subgroups M of the total number N of pulses integrated. The density of probability $P(x, \lambda)$ would therefore be replaced by the sharper, essentially "normal" distribution of M chance variables. That is, the distribution as seen on the screen would be condensed. Of course, such a grouping should have no effect upon the minimum signal visible with brightness modulation.

¹¹ See [5], Section 3.

¹² See [5], Section 12. (Note: Insertion of an electrical feedback loop with $H(\infty) \approx 20$, into a system using deflection modulation and a pulse rate of 3000 sec⁻¹, yielded several db reduction of minimum visible signal. DON, 1963.)

VI. CONCLUSION

While there seems at present no thoroughly satisfactory solution to the moving-target problem in respect to integration before detection of short-wave pulsed carriers, the prospect of integration after detection appears promising, particularly if the indicating means are not encumbered by the contrast limit ϵ of the eye. One such system has already given evidence of the gains to be secured by this means, with a tenfold improvement over the minimum signal visible with conventional receivers.

The concomitant gains to be secured at the transmitting end of the system, obtained by recognition of the appropriate transmitter criterion, should not be overlooked, especially the prospect of enhancing these gains by sacrificing range resolution for the sake of visibility.

As one well familiar with the constant, earnest endeavor to develop, for receivers in this service, tubes yielding the lowest attainable receiver noise factor, perhaps I will be permitted to emphasize to those entrusted with the over-all design and operation of these systems that every twofold improvement in predetector selectivity (U), every fourfold increase in the integration time, every fourfold improvement in the transmitter criterion (Pulse Energy $\times$ Average Power) secures an improvement in visibility equal to that obtained with a 3-db improvement in noise factor.

In a report just received,¹³ A. V. Haeff has described an experimental investigation leading to an empirical formula for minimum visible signal with a conventional receiver and Type-A scan, the result of a statistical study involving a number of observers. The formula is in close functional agreement with the analytical deductions of Section V-B of this report, with perhaps one exception: Haeff's experimental deduction that $\lambda_0 \propto N^{-1/3}$ is at variance with expression (48) unless in his experiments the minimum discernible contrast ϵ is somewhat higher than the 4 per cent estimate used in this report.

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¹³ A. V. Haeff, "Minimum Detectable Radar Signal and its Dependence upon Parameters of Radar Systems," Naval Research Laboratory, Washington, D. C., Confidential Rept. No. RA-3A-208A.

(Note: Subsequently published by A. V. Haeff in *Proc. IRE*, vol. 34, pp. 857-861; November, 1946. In view of the complexity of such experimental work, and the elaborate precautions necessary for high precision, the variance mentioned here appears, in retrospect, much less noteworthy than the broad areas of agreement. DON, 1963.)